Dev Jhawar

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Education

Kalinga Institute of Industrial Technology — Bhubaneswar, India
Bachelor of Technology in Computer Science, CGPA 9.65 — *Aug 2023 – Present*

Experience

Open Source Contributor — Oct 2026 – Present
AI, search and backend projects — *Remote*

- Merged pull requests in open-source AI and Python projects (github.com/DEVJHAWAR11).
- **Pipecat** (voice AI framework): added a time limit to websocket sends, so one stuck connection can no longer freeze the whole voice pipeline. **LiveKit Agents**: fixed speech audio being sent late.
- **txtai** (semantic search): made vector search faster by picking only the top results instead of sorting all scores (about 105 ms to 22 ms per query on 500,000 vectors). Also sped up bulk deletes in the keyword (BM25) index.
- **TruLens** (AI evaluation): added a metric that uses an LLM judge to check if a chatbot stays in its assigned role. The score is calculated by code, not by the LLM.
- **Prefect** (workflow tool with a FastAPI server): fixed a 500 error in the flow run list API when a parameter was a very large integer.

IOT LAB, Technical Community Team — Nov 2024 – Dec 2025
KIIT University — *Bhubaneswar, India*

- Handled sponsor outreach and community work for technical events on campus.

Projects

PolicyLens: RAG Policy Analyzer — Jan 2026
Python, FastAPI, PostgreSQL (Supabase, pgvector), Docker — *GitHub | Live Demo*

- Built a **RAG** (retrieval-augmented generation) app that answers questions about insurance policy PDFs and cites the **exact page and clause**.
- Built the upload pipeline: policies are split by clause, turned into embeddings (**Jina**) and stored in **pgvector** on Supabase. A background worker reads jobs from a PostgreSQL queue, so uploads do not block the API.
- Improved search by combining keyword search (**BM25**) with meaning-based **vector search**, merging both result lists, then re-ranking with a Jina reranker.
- Added **comparison across 2 to 5 policies** with streamed answers (Groq), and an endpoint that creates an automatic summary of a policy.
- **FastAPI** backend (upload, questions, streaming); **Docker**, Render, **GitHub Actions** CI.

GridMind: AI Research Spreadsheet — Oct 2026
Python, FastAPI, LangGraph, Redis, PostgreSQL — *GitHub | Live Demo*

- Built a spreadsheet where **each cell runs its own AI agent** (**LangGraph**). The agent searches the web, reads pages when needed, returns an answer with sources, and searches again if something is missing.
- Built a job queue on **PostgreSQL** (SKIP LOCKED) so several workers share tasks safely, with retries.
- Showed live agent progress in the browser using SSE and **Redis** pub/sub. Added rate limits, retries and a cache.

CaseCompass: Consumer-Court Case Finder — Oct 2026
Python, FastAPI, PostgreSQL, Groq LLM — *GitHub | Live Demo*

- Built a search app over **600+ Indian consumer-court judgments**. A user describes a dispute in plain words and sees how similar cases ended, with the outcome, refund amount and a link to the judgment.
- Used an **LLM** to read each judgment into structured data (JSON) and checked it with code rules. Counts such as "7 of 10 cases" are calculated by code, so the LLM never makes up numbers.
- Improved search by rewriting the story into legal-style wording (tested on 27 samples). **FastAPI** backend with rate limits and a daily token budget, deployed with **Docker** on Render.

Technical Skills

Languages: Python, Java, C/C++, SQL
AI and ML: LLMs, RAG, AI agents, LangChain, LangGraph, embeddings, vector search, PyTorch, Scikit-learn
Backend and Tools: FastAPI, PostgreSQL, Redis, Docker, Git, GitHub Actions, Supabase, MCP
Computer Science: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements

- **Competitive programming:** LeetCode 1000+ problems solved, highest rating 1750 (leetcode.com/u/devjhawar); CodeChef 3 Star, highest 1625; Codeforces Pupil, highest 1250.